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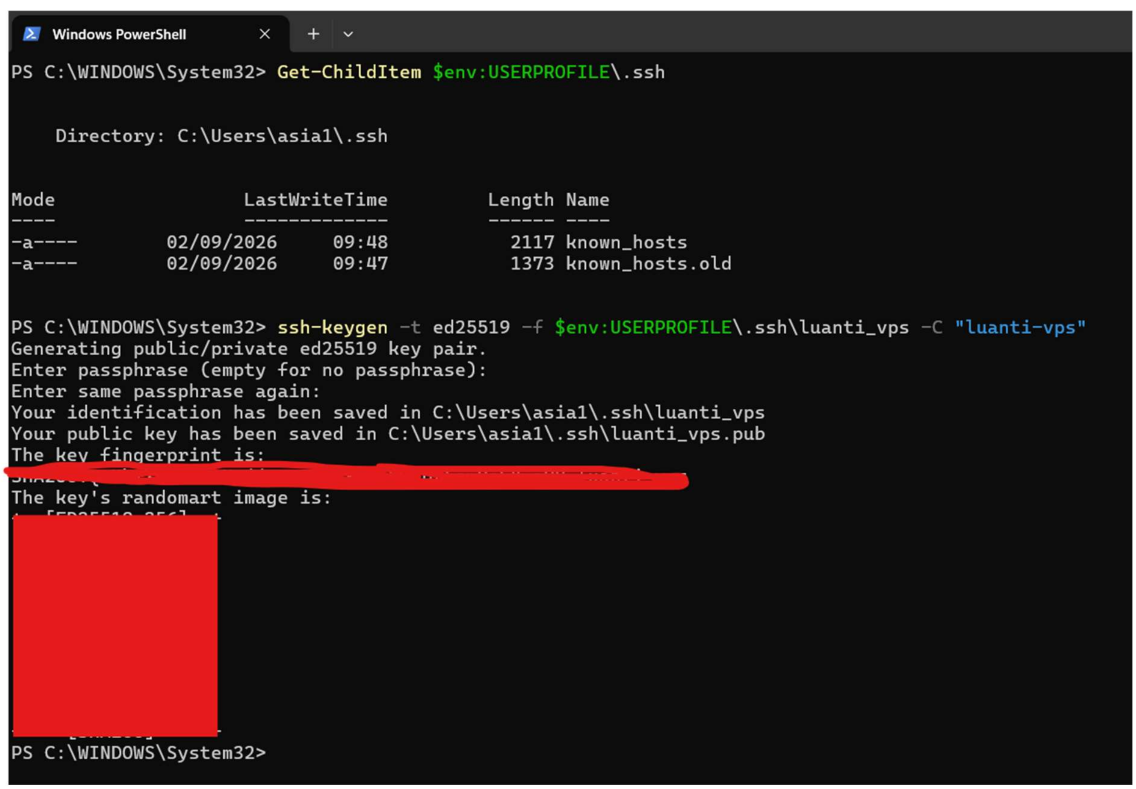
The purpose of this lab is to integrate a public-facing Ubuntu cloud server hosting a Luanti/VoxeLibre game server with a private Splunk-based SOC environment. The server will generate real-world system, authentication, firewall, application and network telemetry, which will be securely forwarded to Splunk for centralised monitoring and analysis. The environment will then be used to perform controlled security testing, develop detections, investigate suspicious activity and demonstrate practical SIEM and incident investigation skills.

The installation and initial configuration of the Luanti/VoxeLibre cloud server are covered in separate documentation. This document focuses on the security monitoring, Splunk integration, and controlled security testing of the deployed server.

Secure SSH Access from Windows (my personal laptop)

Key generation

My first step is to generate a pair of key in order to securely access the cloud server for administrative purposes. The server was initially accessible via SSH using username and password authentication. To improve security, an **Ed25519 SSH key pair** was generated on the Windows administration workstation.



The objective is to replace password-based SSH authentication with **public-key authentication**. The private key remains securely stored on the Windows administration workstation, while the corresponding public key will be added to the Ubuntu VPS. This reduces the risk associated with password-based SSH access, particularly because the server is publicly accessible from the Internet.

Configuring SSH Public Key Authentication on the Ubuntu VPS

The next step was to configure the Ubuntu VPS to accept the previously generated public SSH key. The server was first accessed using the existing password-based SSH authentication.

Once logged in to the server, I added my personal public key using nano:

```
Last login: Wed Sep  2 08:48:12 2026 from 77.102.204.213
ubuntu@vps-1e729caf:~$ mkdir -p ~/.ssh


ubuntu@vps-1e729caf:~$ chmod 600 ~/.ssh/authorized_keys
ubuntu@vps-1e729caf:~$ ls -ld ~/.ssh && ls -l ~/.ssh/authorized_keys
drwx------ 2 ubuntu ubuntu 4096 Sep  2 10:03 /home/ubuntu/.ssh
-rw------- 1 ubuntu ubuntu 92 Sep  2 10:03 /home/ubuntu/.ssh/authorized_keys
ubuntu@vps-1e729caf:~$
```

After successfully adding the key, I opened new session using my private key instead of the password:

```
PS C:\Users\asia1> ssh -i $env:USERPROFILE\.ssh\luanti_vps ubuntu@51.195.218.73
Enter passphrase for key 'C:\Users\asia1\.ssh\luanti_vps':
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-30-generic x86_64)
```

```
Last login: Wed Sep  2 09:59:49 2026 from 77.102.204.213
ubuntu@vps-1e729caf:~$
```

SSH public key authentication was successfully configured and tested. Password authentication was temporarily retained to ensure that other authorised administrators maintained access to the server. It will be reviewed during the server hardening stage.

Preparing the VPS for Splunk Universal Forwarder Installation

Before installing the Splunk Universal Forwarder, basic system information was verified to ensure that the cloud server met the installation requirements and to identify the correct forwarder package. The operating system, CPU architecture, available storage and memory were reviewed.

```
Last login: Wed Sep  2 09:59:49 2026 from 77.102.204.213
ubuntu@vps-1e729caf:~$ hostnamectl
 Static hostname: vps-1e729caf
       Icon name: computer-vm
         Chassis: vm 🖴
      Machine ID: cfcc10ff31564f809df773e2a8aec5fa
         Boot ID: 5f4a9381273c4f558573db04e6307093
  Virtualization: kvm
Operating System: Ubuntu 26.04.1 LTS
          Kernel: Linux 7.0.0-30-generic
    Architecture: x86-64
 Hardware Vendor: OpenStack Foundation
  Hardware Model: OpenStack Nova
Hardware Version: 19.3.2
Firmware Version: 1.16.3-debian-1.16.3-2~bpo12+1
   Firmware Date: Tue 2014-04-01
    Firmware Age: 12y 5month 2d
ubuntu@vps-1e729caf:~$ df -h /
Filesystem      Size  Used Avail Use% Mounted on
/dev/sda1        38G  3.4G   35G   9% /
ubuntu@vps-1e729caf:~$ free -h
               total        used        free      shared  buff/cache   available
Mem:           3.7Gi       950Mi       1.5Gi       2.1Mi       1.6Gi       2.8Gi
Swap:             0B          0B          0B
ubuntu@vps-1e729caf:~$ uname -m
x86_64
ubuntu@vps-1e729caf:~$ |
```

A baseline review of the running services and listening network ports was performed before deploying the Splunk Universal Forwarder. The Luanti Docker container was confirmed to be operational and listening on UDP port 30000, while SSH was available on TCP port 22. This established the server's operational state before introducing Splunk monitoring.

```
ubuntu@vps-1e729caf:~$ docker ps
CONTAINER ID   IMAGE                                COMMAND                  CREATED         STATUS         PORTS
                                                NAMES
0d4915fc8d49   ghcr.io/luanti-org/luanti:latest   "/usr/local/bin/luan…"   18 hours ago   Up 18 hours   0.0.0.0:30000->3
0000/udp, [::]:30000->30000/udp, 30000/tcp   luanti_server
ubuntu@vps-1e729caf:~$ sudo ss -tulpn
Netid        State        Recv-Q        Send-Q                    Local Address:Port                     Peer Address:Port
Process
udp          UNCONN       0             0                               0.0.0.0:30000                         0.0.0.0:*
 users:(("docker-proxy",pid=7616,fd=7))
udp          UNCONN       0             0                            127.0.0.54:53                            0.0.0.0:*
 users:(("systemd-resolve",pid=505,fd=18))
udp          UNCONN       0             0                         127.0.0.53%lo:53                            0.0.0.0:*
 users:(("systemd-resolve",pid=505,fd=16))
udp          UNCONN       0             0                    51.195.218.73%ens3:68                            0.0.0.0:*
 users:(("systemd-network",pid=880,fd=35))
udp          UNCONN       0             0                             127.0.0.1:323                           0.0.0.0:*
 users:(("chronyd",pid=994,fd=4))
udp          UNCONN       0             0                                  [::]:30000                            [::]:*
 users:(("docker-proxy",pid=7622,fd=7))
udp          UNCONN       0             0                                 [::1]:323                              [::]:*
 users:(("chronyd",pid=994,fd=5))
tcp          LISTEN       0             4096                          127.0.0.1:45665                         0.0.0.0:*
 users:(("containerd",pid=2595,fd=13))
tcp          LISTEN       0             4096                            0.0.0.0:22                            0.0.0.0:*
 users:(("sshd",pid=1124,fd=3),("systemd",pid=1,fd=259))
tcp          LISTEN       0             4096                         127.0.0.54:53                            0.0.0.0:*
 users:(("systemd-resolve",pid=505,fd=19))
tcp          LISTEN       0             4096                      127.0.0.53%lo:53                            0.0.0.0:*
 users:(("systemd-resolve",pid=505,fd=17))
tcp          LISTEN       0             4096                               [::]:22                               [::]:*
 users:(("sshd",pid=1124,fd=4),("systemd",pid=1,fd=260))
ubuntu@vps-1e729caf:~$
```

Download Splunk Universal Forwarder

The VPS specifications and active services were checked before installing Splunk Universal Forwarder 10.4.2. The installation was completed and verified successfully.

The forwarder is now installed but not yet connected to the Splunk Server.

```
ubuntu@vps-1e729caf:~$ cd /tmp
ubuntu@vps-1e729caf:/tmp$ wget -O splunkforwarder-10.4.2-33c3bf42cd73-linux-amd64.deb "https://download.splunk.com/produ
cts/universalforwarder/releases/10.4.2/linux/splunkforwarder-10.4.2-33c3bf42cd73-linux-amd64.deb"
--2026-09-02 10:25:15--  https://download.splunk.com/products/universalforwarder/releases/10.4.2/linux/splunkforwarder-1
0.4.2-33c3bf42cd73-linux-amd64.deb
Resolving download.splunk.com (download.splunk.com)... 2600:9000:2117:d400:1d:f9c1:d100:93a1, 2600:9000:2117:9200:1d:f9c
1:d100:93a1, 2600:9000:2117:1a00:1d:f9c1:d100:93a1, ...
Connecting to download.splunk.com (download.splunk.com)|2600:9000:2117:d400:1d:f9c1:d100:93a1|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 88743138 (85M) [binary/octet-stream]
Saving to: ‘splunkforwarder-10.4.2-33c3bf42cd73-linux-amd64.deb’

splunkforwarder-10.4.2-33c3bf 100%[===============================================>]  84.63M  57.6MB/s    in 1.5s

2026-09-02 10:25:17 (57.6 MB/s) - ‘splunkforwarder-10.4.2-33c3bf42cd73-linux-amd64.deb’ saved [88743138/88743138]

ubuntu@vps-1e729caf:/tmp$ ls -lh splunkforwarder-10.4.2-33c3bf42cd73-linux-amd64.deb
-rw-rw-r-- 1 ubuntu ubuntu 85M Jul 28 21:16 splunkforwarder-10.4.2-33c3bf42cd73-linux-amd64.deb
ubuntu@vps-1e729caf:/tmp$
```

```
ubuntu@vps-1e729caf:/tmp$ sudo dpkg -i splunkforwarder-10.4.2-33c3bf42cd73-linux-amd64.deb
Selecting previously unselected package splunkforwarder.
(Reading database ... 133150 files and directories currently installed.)
Preparing to unpack splunkforwarder-10.4.2-33c3bf42cd73-linux-amd64.deb ...
verify that this sytem has all the commands we will require to perform the preflight step
no need to run the splunk-preinstall upgrade check
Unpacking splunkforwarder (10.4.2) ...
Setting up splunkforwarder (10.4.2) ...
find: ‘/opt/splunkforwarder/lib/python3.7/site-packages’: No such file or directory
find: ‘/opt/splunkforwarder/lib/python3.9/site-packages’: No such file or directory
complete
ubuntu@vps-1e729caf:/tmp$ ls -ld /opt/splunkforwarder
drwxr-xr-x 10 splunkfwd splunkfwd 4096 Sep  2 10:27 /opt/splunkforwarder
```

```
ubuntu@vps-1e729caf:/tmp$ sudo -u splunkfwd /opt/splunkforwarder/bin/splunk version
Warning: Attempting to revert the SPLUNK_HOME ownership
Warning: Executing "chown -R splunkfwd:splunkfwd /opt/splunkforwarder"
Splunk General Terms (v4 August 2024)
```

```
Splunk Universal Forwarder 10.4.2 (build 33c3bf42cd73)
ubuntu@vps-1e729caf:/tmp$
```

Secure Connection Between VPS and SOC Lab

For the security purposes, my goal is to separate my internal soc lab from the public internet. In order to do so, I will create a WireGuard VPN tunnel, to securely connect the VPS forwarder to Splunk server.

Public VPS

51.195.218.73

Splunk UF

│

│ VPN / WireGuard

▼

pfSense

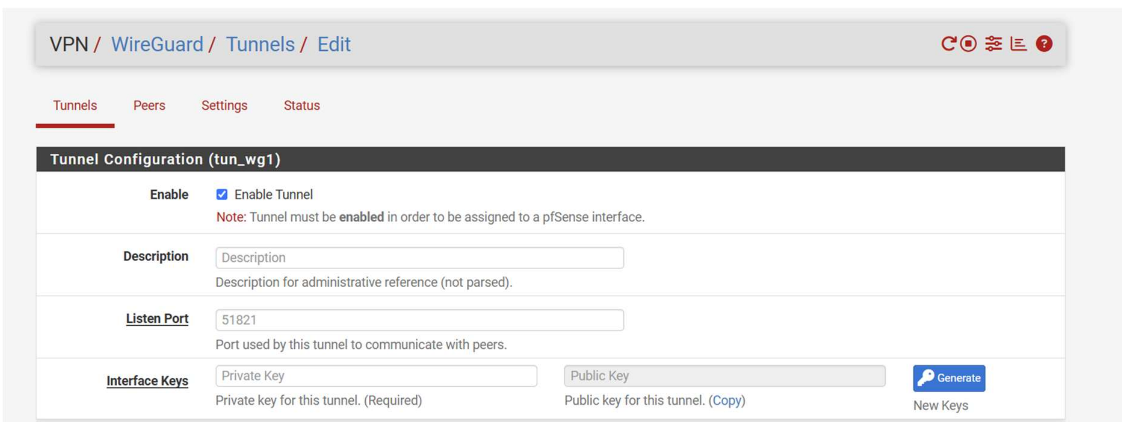
│

│ 192.168.50.0/24

▼

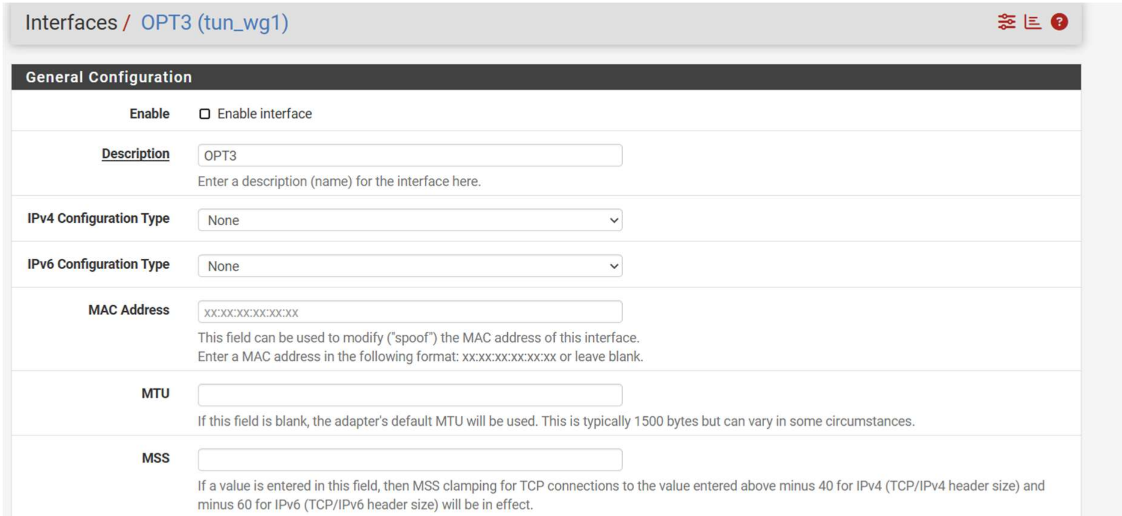
Splunk Server

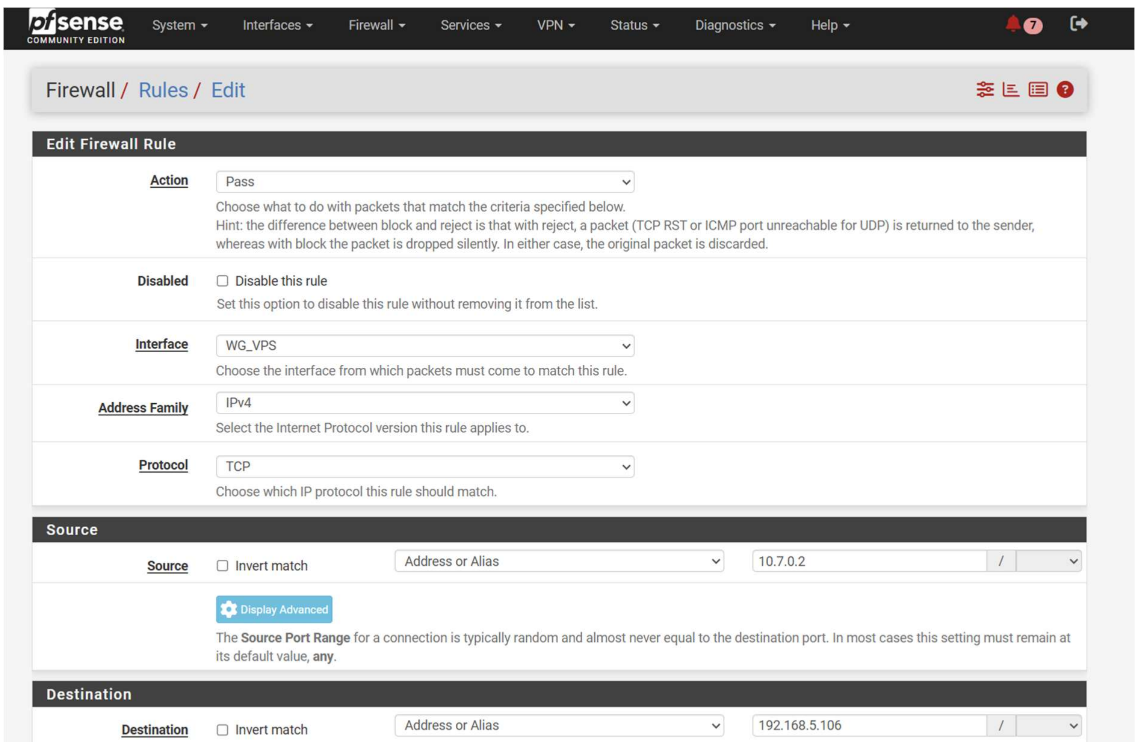
192.168.50.10:9997

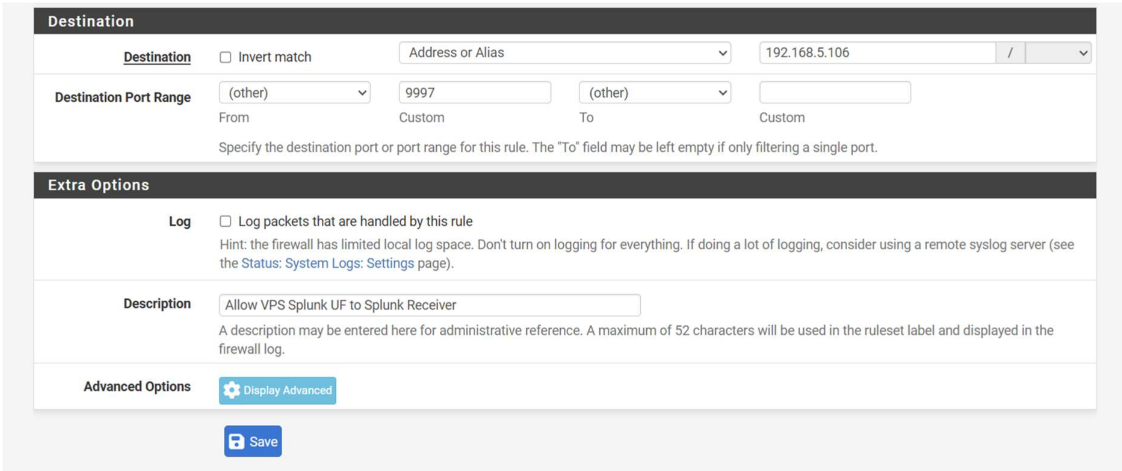


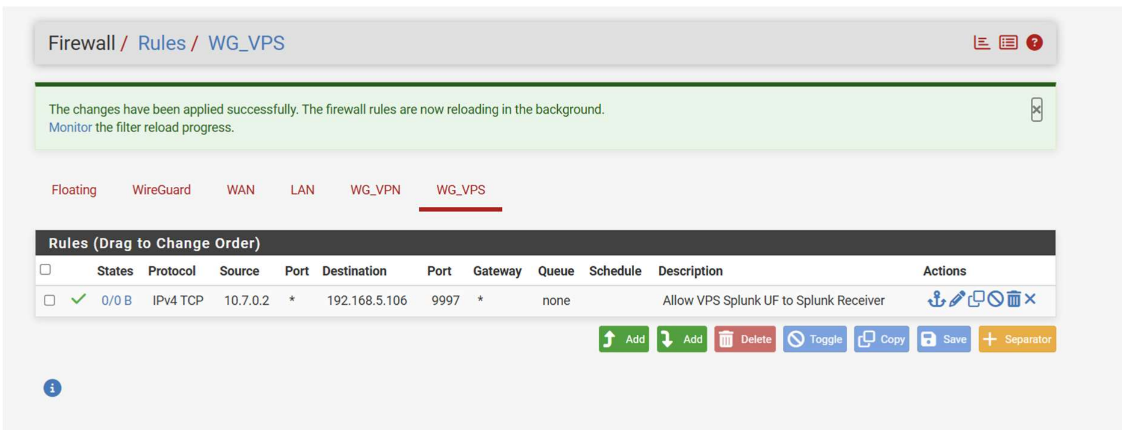
In this step, I created new VPN tunnel, assigned description, listening port and generated a pair of new keys. (information has been added after taking the above screenshot)

Next, I configure the interface to fit my needs and security measures.


Interfaces / OPT3 (tun_wg1)
General Configuration
Enable
Enable interface
Description
OPT3
Enter a description (name) for the interface here.
IPv4 Configuration Type
None
IPv6 Configuration Type
None
MAC Address
xx:xx:xx:xx:xx:xx
This field can be used to modify ("spoof") the MAC address of this interface.
Enter a MAC address in the following format: xx:xx:xx:xx:xx:xx or leave blank.
MTU
If this field is blank, the adapter's default MTU will be used. This is typically 1500 bytes but can vary in some circumstances.
MSS
If a value is entered in this field, then MSS clamping for TCP connections to the value entered above minus 40 for IPv4 (TCP/IPv4 header size) and minus 60 for IPv6 (TCP/IPv6 header size) will be in effect.


pfsense COMMUNITY EDITION
System
Interfaces
Firewall
Services
VPN
Status
Diagnostics
Help
7
Firewall / Rules / Edit
Edit Firewall Rule
Action
Pass
Choose what to do with packets that match the criteria specified below.
Hint: the difference between block and reject is that with reject, a packet (TCP RST or ICMP port unreachable for UDP) is returned to the sender, whereas with block the packet is dropped silently. In either case, the original packet is discarded.
Disabled
Disable this rule
Set this option to disable this rule without removing it from the list.
Interface
WG_VPS
Choose the interface from which packets must come to match this rule.
Address Family
IPv4
Select the Internet Protocol version this rule applies to.
Protocol
TCP
Choose which IP protocol this rule should match.
Source
Source
Invert match
Address or Alias
10.7.0.2
/
Display Advanced
The Source Port Range for a connection is typically random and almost never equal to the destination port. In most cases this setting must remain at its default value, any.
Destination
Destination
Invert match
Address or Alias
192.168.5.106
/





To summarise the above screenshots:

A separate WireGuard tunnel was created on pfSense specifically for the public VPS to keep it isolated from the existing VPN and internal networks.

- Tunnel: **WG_VPS_SPLUNK**
- Interface: **WG_VPS**
- VPN network: **10.7.0.0/30**
- pfSense address: **10.7.0.1**
- VPS address: **10.7.0.2**

A restrictive firewall rule was then created allowing the VPS to communicate **only with the Splunk receiver**:

10.7.0.2 → 192.168.5.106:9997 TCP

All other traffic from the VPS remains blocked by default, protecting the home network and other lab systems.

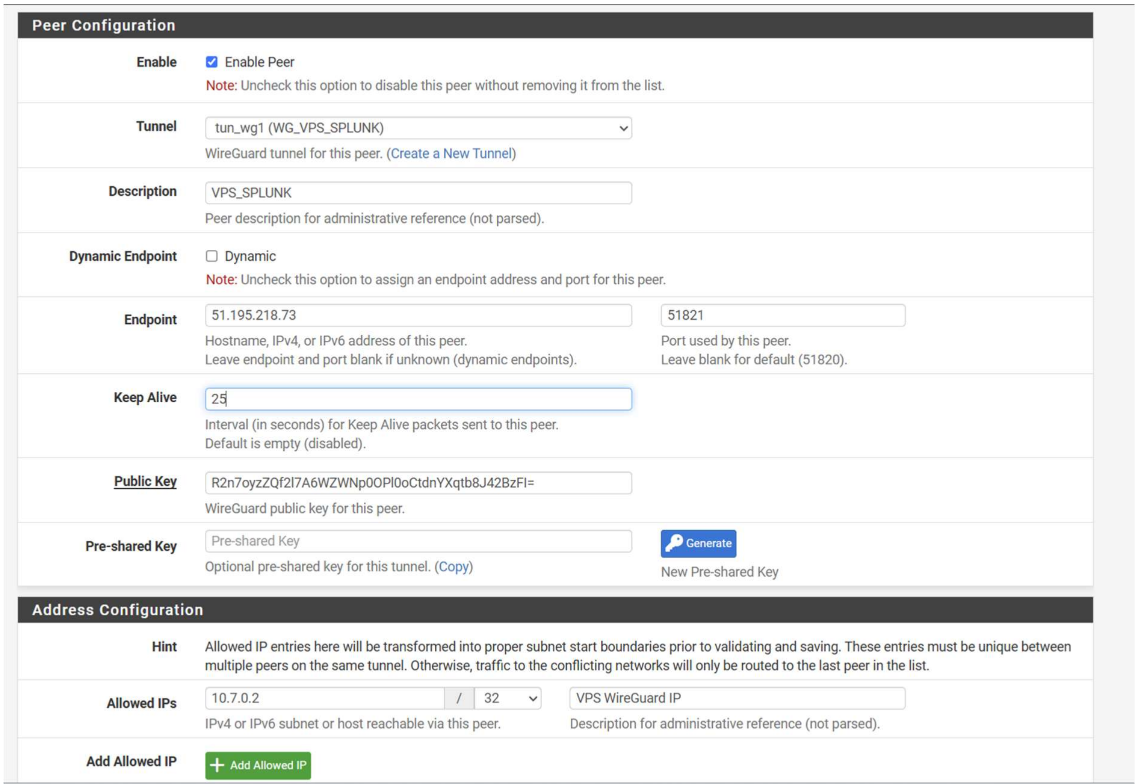
WireGuard Configuration on VPS

In this part, I installed WireGuard on VPS and then generated a pair of keys. The public key of VPS will be then added to the tunnel in the pfSense VPN management.

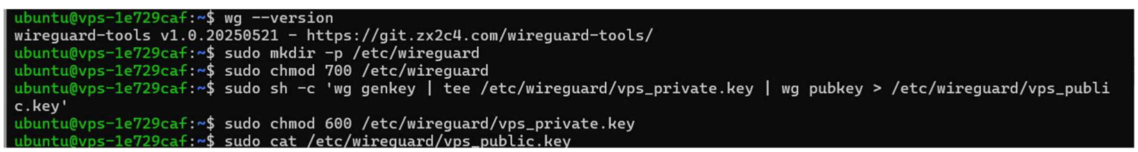
```
ubuntu@vps-1e729caf:~$ sudo apt update
Hit:1 http://nova.clouds.archive.ubuntu.com/ubuntu resolute InRelease
Hit:2 http://security.ubuntu.com/ubuntu resolute-security InRelease
Get:3 http://nova.clouds.archive.ubuntu.com/ubuntu resolute-updates InRelease [137 kB]
Hit:4 http://nova.clouds.archive.ubuntu.com/ubuntu resolute-backports InRelease
Get:5 http://nova.clouds.archive.ubuntu.com/ubuntu resolute-updates/main amd64 Packages [569 kB]
Get:6 http://nova.clouds.archive.ubuntu.com/ubuntu resolute-updates/main Translation-en [137 kB]
Get:7 http://nova.clouds.archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Packages [265 kB]
Get:8 http://nova.clouds.archive.ubuntu.com/ubuntu resolute-updates/universe Translation-en [87.1 kB]
Fetched 1195 kB in 1s (1030 kB/s)
1 package can be upgraded. Run 'apt list --upgradable' to see it.
ubuntu@vps-1e729caf:~$ sudo apt install wireguard-tools
The following package was automatically installed and is no longer required:
  pollinate
Use 'sudo apt autoremove' to remove it.

Installing:
  wireguard-tools

Summary:
  Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 1
  Download size: 89.6 kB
  Space needed: 327 kB / 36.5 GB available

Get:1 http://nova.clouds.archive.ubuntu.com/ubuntu resolute/main amd64 wireguard-tools amd64 1.0.20250521-1ubuntu1 [89.6
 kB]
```

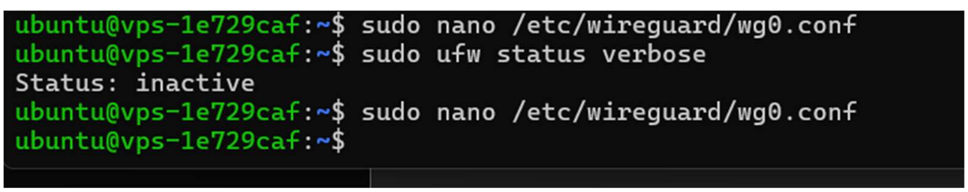
```
ubuntu@vps-1e729caf:~$ wg --version
wireguard-tools v1.0.20250521 - https://git.zx2c4.com/wireguard-tools/
ubuntu@vps-1e729caf:~$ sudo mkdir -p /etc/wireguard
ubuntu@vps-1e729caf:~$ sudo chmod 700 /etc/wireguard
ubuntu@vps-1e729caf:~$ sudo sh -c 'wg genkey | tee /etc/wireguard/vps_private.key | wg pubkey > /etc/wireguard/vps_publi
c.key'
ubuntu@vps-1e729caf:~$ sudo chmod 600 /etc/wireguard/vps_private.key
ubuntu@vps-1e729caf:~$ sudo cat /etc/wireguard/vps_public.key
R2n7oyzZQf2l7A6WZWNp0OPl0oCtdnYXqtb8J42BzFI=
ubuntu@vps-1e729caf:~$
```

Below, I added a peer to the VPS Tunnel, using the VPS’s public key.



Now I will add pfSense public key for WG_VPS_SPLUNK.





Finally, the VPN tunnel is active and working as expected.

```
ubuntu@vps-1e729caf:~$ sudo wg-quick up wg0
[#] ip link add dev wg0 type wireguard
[#] wg setconf wg0 /dev/fd/63
[#] ip -4 address add 10.7.0.2/30 dev wg0
[#] ip link set mtu 1420 up dev wg0
[#] ip -4 route add 192.168.5.106/32 dev wg0
ubuntu@vps-1e729caf:~$ sudo wg
interface: wg0
  public key: R2n7oyzZQf2l7A6WZWNp0OPl0oCtdnYXqtb8J42BzFI=
  private key: (hidden)
  listening port: 51821

peer: h17mAgwMpi3f2YpEq+88J2dZqBx0WSIpfAX9seytUHg=
  endpoint: xxx
  allowed ips: 10.7.0.1/32, 192.168.5.106/32
  latest handshake: 7 seconds ago
  transfer: 180 B received, 92 B sent
ubuntu@vps-1e729caf:~$
```

```
ubuntu@vps-1e729caf:~$ nc -vz -w 3 192.168.5.106 9997
Connection to 192.168.5.106 9997 port [tcp/*] succeeded!
ubuntu@vps-1e729caf:~$
```

Tunnel configured successfully.

Forwarder Configuration

Forwarding enabled and receiver specified as below:

```
ubuntu@vps-1e729caf:~$ sudo -u splunkfwd /opt/splunkforwarder/bin/splunk add forward-server 192.168.5.106:9997
Warning: Attempting to revert the SPLUNK_HOME ownership
Warning: Executing "chown -R splunkfwd:splunkfwd /opt/splunkforwarder"
tcp_conn_open_afux ossocket_connect failed with No such file or directory
tcp_conn_open_afux ossocket_connect failed with No such file or directory
tcp_conn_open_afux ossocket_connect failed with No such file or directory
Added forwarding to: 192.168.5.106:9997.
```

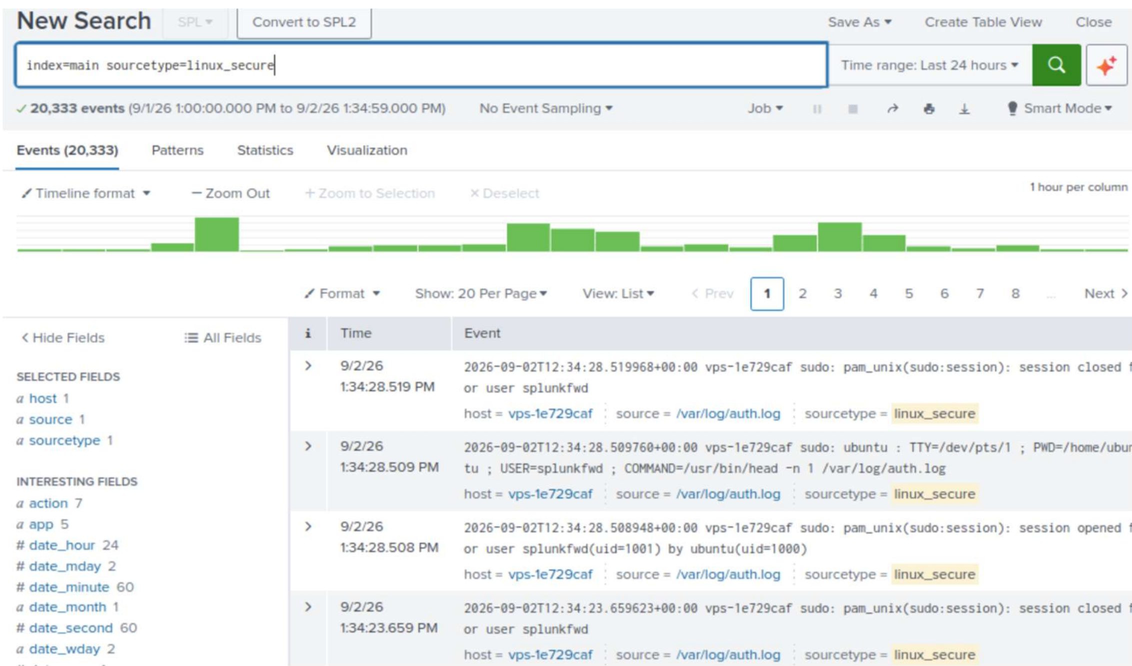
```
ubuntu@vps-1e729caf:~$ sudo -u splunkfwd /opt/splunkforwarder/bin/splunk list forward-server
Warning: Attempting to revert the SPLUNK_HOME ownership
Warning: Executing "chown -R splunkfwd:splunkfwd /opt/splunkforwarder"
Splunk username: admin
Password:
Login failed
ubuntu@vps-1e729caf:~$ sudo -u splunkfwd /opt/splunkforwarder/bin/splunk list forward-server
Warning: Attempting to revert the SPLUNK_HOME ownership
Warning: Executing "chown -R splunkfwd:splunkfwd /opt/splunkforwarder"
Splunk username: admin
Password:
Active forwards:
        192.168.5.106:9997
Configured but inactive forwards:
        None
ubuntu@vps-1e729caf:~$
```

```
ubuntu@vps-1e729caf:~$ sudo mkdir -p /var/log/luanti
ubuntu@vps-1e729caf:~$ sudo chown root:splunkfwd /var/log/luanti
ubuntu@vps-1e729caf:~$ sudo chmod 750 /var/log/luanti
ubuntu@vps-1e729caf:~$ ls -ld /var/log/luanti
drwxr-x--- 2 root splunkfwd 4096 Sep  2 12:20 /var/log/luanti
ubuntu@vps-1e729caf:~$
```

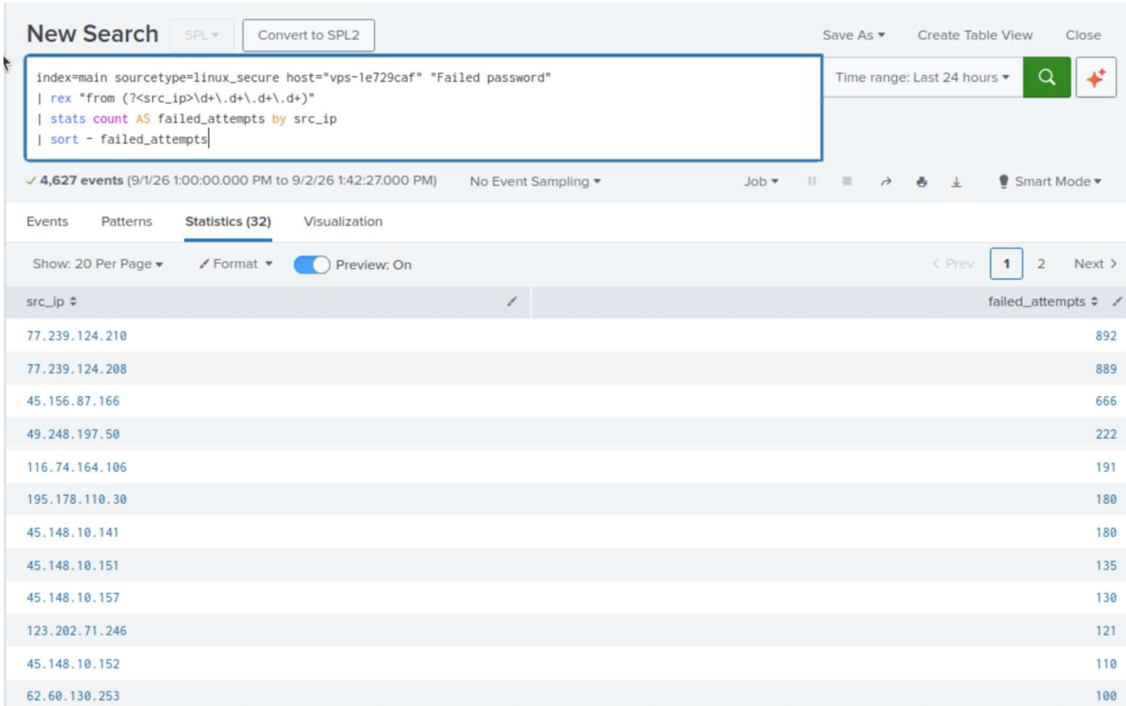
Adding logs to be forwarded to Splunk:

```
ubuntu@vps-1e729caf:~$ sudo -u splunkfwd /opt/splunkforwarder/bin/splunk add monitor /var/log/auth.log -sourcetype linux
_secure -index main
Warning: Attempting to revert the SPLUNK_HOME ownership
Warning: Executing "chown -R splunkfwd:splunkfwd /opt/splunkforwarder"
Added monitor of '/var/log/auth.log'.
ubuntu@vps-1e729caf:~$
```

Logs successfully forwarded to Splunk Server, which can be confirmed in Splunk Search:



As we can see below, in the last 24 hours, server as heavily attacked with failed loggings attempts. All the logs will be analysed in a different document.



Lab Summary

A secure monitoring connection was successfully established between the public Luanti VPS and the internal Splunk environment using a dedicated WireGuard tunnel. The VPS was isolated from the rest of the home lab through pfSense firewall rules, allowing communication only with the Splunk receiver on TCP port 9997.

Splunk Universal Forwarder was configured on the VPS and successfully connected to the Splunk server through the encrypted tunnel. Rather than forwarding high-volume gameplay logs, /var/log/auth.log was selected to provide security-relevant information such as SSH authentication failures, invalid user attempts, source IP addresses and privileged activity.

Initial analysis in Splunk immediately identified significant automated SSH activity against the Internet-facing VPS. Within a 24-hour period, **4,627 failed authentication attempts from 32 unique source IP addresses** were recorded. The most active individual source generated **892 failed login attempts**.

This integration provides the home SOC lab with real-world security telemetry from an Internet-facing system while maintaining strict separation between the public VPS and the internal network. The collected data can now be used to develop detections, alerts and dashboards for activity such as SSH brute-force attacks.